

WEEK 14

Week-14-Structures and Unions: Attempt review | REC-CIS - Personal - Microsoft Edge

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REC-CIS

Correct
Flag question

You are transporting some boxes through a tunnel, where each box is a parallelepiped, and is characterized by its length, width and height.

The height of the tunnel **41** feet and the width can be assumed to be infinite. A box can be carried through the tunnel only if its height is strictly less than the tunnel's height. Find the volume of each box that can be successfully transported to the other end of the tunnel. Note: Boxes cannot be rotated.

Input Format

The first line contains a single integer n , denoting the number of boxes.
 n lines follow with three integers on each separated by single spaces - $length_i$, $width_i$ and $height_i$ which are length, width and height in feet of the i -th box.

Constraints

$1 \leq n \leq 100$
 $1 \leq length_i, width_i, height_i \leq 100$

Output Format

For every box from the input which has a height less than **41** feet, print its volume in a separate line.

Sample Input 0

4

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Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 struct box{
3     int l,w,b;
4 };
5
6 int main()
7 {
8     int n;
9     scanf("%d",&n);
10    struct box arr[n];
11    for(int i=0;i<n;i++){
12        scanf("%d %d %d",&arr[i].l,&arr[i].w,&arr[i].b);
13    }
14    for(int i=0;i<n;i++){
15
16
17        if(arr[i].b<41){
18            printf("%d\n",arr[i].l*arr[i].w*arr[i].b);
19        }
20    }
21 }
22
23 }
```

	Input	Expected	Got	
✓	4	125	125	✓
	5 5 5	80	80	
	1 2 40			
	10 5 41			
	7 2 42			

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REC-CIS

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You are given n triangles, specifically, their sides a , b and c . Print them in the same style but sorted by their areas from the smallest one to the largest one. It is guaranteed that all the areas are different.

The best way to calculate a volume of the triangle with sides a , b and c is Heron's formula:

$$S = \sqrt{p \cdot (p - a) \cdot (p - b) \cdot (p - c)} \text{ where } p = (a + b + c) / 2.$$

Input Format

First line of each test file contains a single integer n . n lines follow with a , b and c on each separated by single spaces.

Constraints

$1 \leq n \leq 100$
 $1 \leq a, b, c \leq 70$
 $a + b > c$, $a + c > b$ and $b + c > a$

Output Format

Print exactly n lines. On each line print 3 integers separated by single spaces, which are a , b and c .

Sample Input 0

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```
1 #include<stdio.h>
2 struct sides{
3     int a,b,c;
4 };
5
6 int vol(int a,int b,int c){
7     return a*b*c;
8 }
9 int main()
10 {
11     int n;
12     scanf("%d",&n);
13     struct sides s[n];
14     for(int i=0;i<n;i++){
15         scanf("%d %d %d",&s[i].a,&s[i].b,&s[i].c);
16     }
17     for(int i=0;i<n;i++){
18         for(int j=i+1;j<n;j++){
19             if(vol(s[i].a,s[i].b,s[i].c)<vol(s[j].a,s[j].b,s[j].c)){
20                 struct sides temp=s[j];
21                 s[j]=s[i];
22                 s[i]=temp;
23             }
24         }
25     }
26     for(int i=0;i<n;i++){
27         printf("%d %d %d\n",s[i].a,s[i].b,s[i].c);
28     }
29 }
30 }
```

	Input	Expected	Got	
✓	3	3 4 5	3 4 5	✓

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REC-CIS

```
14 for(int i=0;i<n;i++){
15     scanf("%d %d %d",&s[i].a,&s[i].b,&s[i].c);
16
17 }for(int i=0;i<n;i++){
18     for(int j=0;j<n;j++){
19         if(i!=j){
20             if(vol(s[i].a,s[i].b,s[i].c)<vol(s[j].a,s[j].b,s[j].c)){
21                 struct sides temp=s[j];
22                 s[j]=s[i];
23                 s[i]=temp;
24             }
25         }
26     }
27 }for(int i=0;i<n;i++){
28     printf("%d %d\n",s[i].a,s[i].b,s[i].c);
29 }
30 }
```

	Input	Expected	Got	
✓	3	3 4 5	3 4 5	✓
	7 24 25	5 12 13	5 12 13	
	5 12 13	7 24 25	7 24 25	
	3 4 5			

Passed all tests! ✓

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12-01-2025